

Pseudocode

Question 1

```
DECLARE N, Rate, nights : INTEGER  
DECLARE DECLARE season, Roomtype : STRING  
DECLARE DECLARE Revenue, Discount, Final : REAL  
BEGIN  
  let  
Input N ← input from user.  
  Revenue = 0
```

```
FOR Count ← 1 TO N :
```

```
  let season ← input from user
```

```
  let Roomtype ← input from user
```

```
  let nights ← input from user
```

```
  IF season = "Peak" THEN
```

```
    IF Roomtype = "Standard" THEN
```

```
      Rate ← 5000
```

```
    ELSE IF Roomtype = "Deluxe" THEN
```

```
      Rate ← 8000
```

```
    ELSE
```

```
      Rate ← 12000
```

```
    ENDIF
```

```
  ELSE
```

```
    IF Roomtype = "Standard" THEN
```

```
      Rate ← 3000
```

```
    ELSE IF Roomtype = "Deluxe" THEN
```

```
      Rate ← 5000
```

```
    ELSE
```

```
      Rate ← 8000
```

```
    ENDIF
```

```
  ENDIF
```

```
price  $\leftarrow$  Rate  $\times$  nights  
IF nights  $>$  7 THEN  
    Discount  $\leftarrow$  (price  $\times$  15) / 100  
ELSE  
    Discount  $\leftarrow$  0
```

```
ENDIF
```

```
Final  $\leftarrow$  price - Discount  
Revenue  $\leftarrow$  Revenue + Final
```

```
OUTPUT Final.
```

```
ENDFOR
```

```
OUTPUT Revenue
```

```
END.
```

Question 2

```
DECLARE N, currentfloor, Count, requestedfloor: INTEGER  
DECLARE
```

```
BEGIN
```

```
Let N  $\leftarrow$  input from user
```

```
Currentfloor  $\leftarrow$  0
```

```
FOR Count  $\leftarrow$  1 TO N :
```

```
    let requestedfloor  $\leftarrow$  input from user
```

```
    IF requestedfloor  $>$  Currentfloor THEN
```

```
        OUTPUT "Moving up"
```

```
    ELSEIF requestedfloor  $<$  Currentfloor THEN
```

```
        OUTPUT "Moving Down"
```

```

ELSE
    OUTPUT "Doors Opening"
ENDIF
Currentfloor = requestedfloor
ENDFOR
END

```

Question 3

```

DECLARE N, Count, Sum, i, marks, average : INTEGER
DECLARE average Flag : BOOLEAN
DECLARE result : STRING
BEGIN
    let N ← input from user
    FOR Count ← 1 TO N
        Sum ← 0
        Flag ← FALSE
        FOR i ← 1 TO 5
            let marks ← input from user
            Sum ← Sum + marks
            IF marks < 33 THEN
                Flag ← TRUE
            ENDIF
        ENDFOR
        average ← Sum / 5
        IF average >= 80 THEN
            result ← "Distinction"
        ELSEIF average >= 60 THEN
            result ← "Pass"

```


ELSE

result $\leftarrow$ "FAIL"

ENDIF

IF Flag = TRUE THEN

result $\leftarrow$ "Fail - Subject Deficiency"

ENDIF

OUTPUT average

OUTPUT result

ENDFOR

END

Question 4

DECLARE Quantity : INTEGER.

DECLARE price, tax, discount, Total, ~~Discount~~, final ~~Amount~~ REAL

BEGIN

let Quantity $\leftarrow$ input from user

let price $\leftarrow$ input from user

let tax $\leftarrow$ input from user

let discount $\leftarrow$ input from user.

IF Quantity = Invalid THEN

OUTPUT "Invalid Quantity"

Break

ELSE IF price = Invalid THEN

OUTPUT "Invalid price"

Break

ELSE IF discount = Invalid THEN

OUTPUT "Invalid discount"

Break

ELSE IF tax = Invalid THEN

OUTPUT "Invalid Tax"

Break

ELSE

Total $\leftarrow$ Quantity $\times$ price

DisAmount $\leftarrow$ Total $- ((\text{Total} \times \text{discount}) / 100)$

Final $\leftarrow$ DisAmount $+ ((\text{DisAmount} \times \text{tax}) / 100)$

ENDIF

OUTPUT Total

OUTPUT DisAmount

OUTPUT Final

END

Question 5

DECLARE N, ZoneA, ZoneB, ZoneC, Accept, Reject, Car,
Bike, Van, Count : INTEGER
DECLARE Assigned : BOOLEAN

BEGIN

let N $\leftarrow$ input from user

ZoneA $\leftarrow$ 0

ZoneB $\leftarrow$ 0

ZoneC $\leftarrow$ 0

Accept $\leftarrow$ 0

Reject $\leftarrow$ 0

Car $\leftarrow$ 0

Bike $\leftarrow$ 0

Van $\leftarrow$ 0


```

FOR Count ← 1 TO N
  let vehicle ← input from user
  let category ← input from user
  let permit ← input from user
  let emergency ← input from user
  Flag ← FALSE
  WHILE (vehicle = "V" OR vehicle = "C" OR
vehicle = "B") AND (category = "F" OR category
= "S" OR category = "G") AND (permit =
"N" OR permit = "Y") :
    Assigned ← FALSE
    Flag ← TRUE

```

ENDWHILE

IF Flag = TRUE THEN

IF permit = "N" AND emergency = "N" THEN

OUTPUT "Rejected & Invalid Permit"

Reject ← Reject + 1

ELSE IF category = "F" THEN

IF vehicle = "V" THEN

IF ZoneA < 20 THEN

ZoneA ← ZoneA + 2

Assigned ← TRUE

Assign Zone A

ENDIF

ELSE

IF ZoneA < 20 THEN

Assign ZoneA

ZoneA ← ZoneA + 1

Assigned ← TRUE

ENDIF

ENDIF

ELSE IF category = "S" THEN
IF vehicle = "V" THEN
IF zoneC < 15 THEN
Assign zoneC
- zoneC $\leftarrow$ zoneC + 2
Assigned $\leftarrow$ TRUE
ENDIF

ELSE
IF zoneB < 40 THEN
Assign zoneB
zoneB $\leftarrow$ zoneB + 1
Assigned $\leftarrow$ TRUE
ENDIF

ENDIF
ELSE IF category = "G" THEN
IF vehicle = "V" THEN
IF zoneC < 14 THEN
Assign zoneC
zoneC $\leftarrow$ zoneC + 2
Assigned $\leftarrow$ TRUE
ENDIF

ELSE
IF zoneC < 15 THEN
Assign zoneC
zoneC $\leftarrow$ zoneC + 1
Assigned $\leftarrow$ TRUE
ENDIF

ENDIF



Pseudocode

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IF Assigned = TRUE THEN

Accept $\leftarrow$ Accept + 1

IF vehicle = "C" THEN

Car $\leftarrow$ Car + 1

ELSEIF vehicle = "B" THEN

Bike $\leftarrow$ Bike + 1

ELSE

Van $\leftarrow$ Van + 1

ENDIF

OUTPUT Assigned zone

OUTPUT Remaining

ELSE

Reject $\leftarrow$ Reject + 1

OUTPUT "Rejected - No available suitable zone"

ENDIF

ENDIF

ENDFOR

OUTPUT Accept

OUTPUT Reject

OUTPUT Car

OUTPUT Bike

OUTPUT Van

OUTPUT ZoneA, ZoneB, ZoneC

IF ZoneA = 20 AND ZoneB = 40 AND ZoneC = 15 THEN

```

        OUTPUT "Entire parking is full"
    ELSE
        OUTPUT "Parking is not full"
    ENDIF
END.

```

Question 6

```

DECLARE CurrentBattery, RequiredBattery, Time, Rate, Discount, Cost; INT
DECLARE StationAvailability, vehicle, Disability, Membership; CHAR
DECLARE priority, peak, warning; STRING
BEGIN
    let vehicle vehicle ← input from user
    let CurrentBattery ← input from user
    let RequiredBattery ← input from user
    let Hours ← input from user
    let Time ← input from user
    let Membership ← input from user
    let Disability ← input from user
    let StationAvailability ← input from user.

```

```

IF StationAvailability = "N" THEN
    IF vehicle = "H" THEN
        OUTPUT "Charging unavailable, parking only"
    ELSE
        OUTPUT "No charging slot available"
    ENDIF
ELSE

```


IF vehicle = "H" AND Current Battery ≥ 40 THEN
~~OUTPUT~~ OUTPUT "Vehicle not eligible for charging"

ELSE

Charging $\leftarrow$ Required Battery - Current Battery

IF Required Battery $\leq$ Current Battery THEN

OUTPUT "No charging required"

ELSE

IF Current Battery ≤ 15 AND

Required Battery ≥ 80 THEN

priority $\leftarrow$ "Emergency charging"

ELSE IF Disability = "Y" OR

(Membership = "Y" AND Current Battery ≤ 30) THEN

priority $\leftarrow$ "Charging priority"

ELSE

priority $\leftarrow$ "Normal"

ENDIF

IF ~~current~~ Time < 17 OR Time > 22 THEN

Rate $\leftarrow 35$

peak $\leftarrow$ "Off-Peak"

IF Membership = "Y" AND

priority $>$ "Emergency" THEN

Discount $\leftarrow 20$

ELSE

Discount $\leftarrow 0$

ELSE ENDF

(Part B)

Rate $\leftarrow$ 50

Peak $\leftarrow$ "Peak"

Discount $\leftarrow$ 10

ENDIF

Cost $\leftarrow$ Rate + Discount

IF Hours $\leq$ 2 THEN

parking $\leftarrow$ 200

ELSEIF Hours $\leq$ 5 THEN

parking $\leftarrow$ 400

ELSE

parking $\leftarrow$ 700

ENDIF

IF Disability = "Y" THEN

parking $\leftarrow$ 0

ELSEIF Membership = "Y" THEN

parking $\leftarrow$ parking - $(20 \times \text{parking} / 100)$

ENDIF

Total $\leftarrow$ parking + Cost

IF Hours $>$ 8 THEN

warning $\leftarrow$ "Long stay warning: kindly
relocate your vehicle after charging".

ELSE

warning $\leftarrow$ "Standard parking duration".

ENDIF

OUTPUT vehicle

OUTPUT Current Battery, Required Battery

OUTPUT priority

OUTPUT Peak

OUTPUT Cost

OUTPUT parking

OUTPUT Final

OUTPUT warning

ENDIF

ENDIF

ENDIF

END.